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Adjustable and stowable workstation assembly

Abstract

A workstation holding one or more electronic devices, e.g., a display, keyboard, or the like, can be mounted on a structure, e.g., a wall, using an articulating arm assembly including one or more arms. The one or more arms can be rotatably coupled to the structure on one end and a work surface can be coupled to the other end. At least one of the one or more arms can be height adjustable. The workstation can rotate about a horizontal axis between a use configuration and a storage configuration. The one or more arms can be configured to translate the workstation between a contracted orientation and extended orientation. The height adjustable arm can include a counterbalance mechanism to counter a weight of the one or more components coupled to the workstation. A tension of the one or more springs contained in the counterbalance mechanism can be adjustable.

Inventors: Barros; Michael P. (Minnetonka, MN), Wong; Thiem Chan Duong (Brooklyn Park, MN), Londborg; Linnea (Minneapolis, MN), Laureijs; Andrew P. (Saint Paul, MN), Theis; John William (St. Paul, MN)

Applicant: Ergotron, Inc. (St. Paul, MN)

Family ID: 86903615

Assignee: Ergotron, Inc. (St. Paul, MN)

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Primary Examiner: Epps; Todd M

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

CLAIM OF PRIORITY (1) This patent application is a continuation of U.S. patent application Ser. No. 18/276,515, entitled “ADJUSTABLE AND STOWABLE WORKSTATION ASSEMBLY,” filed Aug. 9, 2023, which is a U.S. NSPCT of PCT Application Serial Number. PCT/US2022/079953, entitled “ADJUSTABLE AND STOWABLE WORKSTATION ASSEMBLY,” filed Nov. 16, 2022, and published as WO 2023/122398 on Jun. 29, 2023, which claims the benefit of priority of Barros, et al. U.S. Provisional Patent Application Ser. No. 63/292,972, entitled “ADJUSTABLE AND STOWABLE WORKSTATION ASSEMBLY,” filed on Dec. 22, 2021, which are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) This document pertains generally, but not by way of limitation, to a workstation having a keyboard tray and a display mount coupled to a structure, more specifically an articulating arm assembly for adjusting the position of the workstation relative to the structure.

BACKGROUND

(2) Workstations holding one or more electronic equipment (e.g., one or more displays, an input device, or the like) are commonly used in modern offices, hospitals, industrial facilities, or other places. A workstation can include an articulating arm assembly, a worksurface/keyboard tray assembly, and a display mount assembly. The workstation can be mounted on a structure (e.g., a wall, a desk, or the like). The arm assembly can provide either height adjustment or articulation or both for the electronic equipment coupled to the workstation. A user of the workstation can easily adjust an orientation of the electronic equipment to fit changing posture of the user throughout the day using the arm assembly.

(3) In order to reduce the space occupied by the workstation when it is not in active use, the workstation can be generally stored in a stowed orientation where the arm assembly can be collapsed close to the mounting location on the structure, and the worksurface/keyboard tray assembly can be folded towards the mounting location. During the use of the workstation, the display and the input device can be pulled away from the mounting location in an extended orientation. In the extended orientation, the workstation can be more accessible, and it can allow the user to interact with user's environment (e.g., to interact with a patient in a hospital room, or the like). A workstation that can be easily manipulated between the stowed orientation and the extended orientation will improve user satisfaction, and it will be adjusted more frequently to provide a better ergonomic positioning for the user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings are illustrative of embodiments of the present invention and therefore do not limit the scope of the invention. The drawings are not to scale and are intended for use in conjunction with the explanations in the following detailed description. Like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

(2) FIG. 1 is an isometric view of an example of a workstation having an arm assembly in an extended orientation.

(3) FIG. 2 is an isometric view of the workstation of FIG. 1 in a stowed orientation.

(4) FIG. 3 is a side view of an example of a workstation having a single height adjustable arm.

(5) FIG. 4A is a partial exploded view of the platform and the arm assembly of FIGS. 1-2.

(6) FIG. 4B is a partial exploded view of the platform and the arm assembly of FIGS. 1 and 3.

(7) FIG. 5 is an isometric view of the rear end of the platform of FIG. 1.

(8) FIG. 6 illustrates the workstation of FIG. 1 in a stowed orientation of the platform.

(9) FIG. 7 is a rear isometric view of the platform of FIG. 4A.

(10) FIG. 8 is an isometric view of an example of an interface assembly in a folded orientation.

(11) FIG. 9 is an isometric view of the interface assembly of FIG. 8 in an extended orientation.

(12) FIG. 10 is an isometric view of an example of a carrier block of FIG. 8.

(13) FIG. 11 is an isometric view of an example of a hinge bracket of FIG. 8.

(14) FIG. 12 is an isometric view of an example of a spring assembly.

(15) FIG. 13 is a bottom isometric view of an example of the interface assembly of FIG. 9 in extended orientation.

- (16) FIG. 14 is a bottom isometric view of another example of the interface assembly of FIG. 9 in extended orientation.
- (17) FIG. 15 is a cross-sectional view of the interface assembly of FIG. 9 in an unlocked configuration.
- (18) FIG. 16 is a cross-sectional view of the interface assembly of FIG. 9 in a locked configuration.
- (19) FIG. 17 is a side view of the second arm of FIG. 1.
- (20) FIG. 18 is a cross-sectional side view of the second arm of FIG. 17.
- (21) FIG. 19 is an enlarged cross-sectional view of an example of the tension adjustment assembly of FIG. 18.
- (22) FIG. 20 is an exploded view of the tension adjustment assembly of FIG. 19.
- (23) FIG. 21 is a bottom isometric view of an example of the lower link arm of FIG. 17.
- (24) FIG. 22 is an isometric view of an example of a housing of FIG. 19.
- (25) FIG. 23 is an isometric view of a gear assembly according to an example configuration of the current disclosure.

OVERVIEW

(26) This disclosure describes a workstation having an articulating arm assembly to support one or more electronic devices including, but not limited to, electronic displays, keyboards, laptops and other computing equipment, medical equipment, scanners, and the like. The workstation can be mounted on a structure including any fixed or mobile support surface (e.g., a wall, a desk, a pole, a cart, or the like). The articulating arm assembly can allow the workstation to translate between a range of positions between stowed and extended orientations.

DETAILED DESCRIPTION

(27) The following detailed description is exemplary in nature and is not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the following description provides some practical illustrations for implementing exemplary embodiments of the present invention. Examples of constructions, materials, dimensions, and manufacturing processes are provided for selected elements, and all other elements employ that which is known to those of ordinary skill in the field of the invention. Those skilled in the art will recognize that many of the noted examples have a variety of suitable alternatives.

(28) FIGS. 1-2 are isometric views of a workstation 100 including an arm assembly (e.g., the arm assembly 110A of FIG. 1) in an extended orientation and in a stowed orientation, respectively, according to some example configurations of the current disclosure. The arm assembly 110A can be coupled to a structure interface 112 on one end, and a mount assembly 120A can be coupled to the arm assembly 110A on the other end. The arm assembly 110A can be configured to translate the mount assembly 120A relative to the structure interface 112. In some example configurations, the arm assembly 110A can include a first arm 114 (e.g., an extension arm) and a second arm 116 rotatably coupled to the first arm 114. The arm assembly 110A can translate the mount assembly 120A relative to the structure interface 112. In some example configurations, the first arm 114 and the second arm 116 can be configured to cooperate to translate the mount assembly in a horizontal direction between the extended orientation and the stowed orientation. In some example configurations, the second arm 116 can be height adjustable. The second arm 116 can translate the mount assembly in a vertical direction relative to the structure interface 112.

(29) The structure interface 112 can be mounted on a fixed or mobile structure (e.g., a wall, a desk, a cart, a pole, or the like). In some example configurations, the structure interface 112 can include an interface bracket. The interface bracket can be directly mounted on a structure. In other configurations, the structure interface 112 can include an interface assembly. The interface assembly can include one or more components to mount the arm assembly 110A on a structure. For example, the interface assembly can include an interface bracket and a wall track. The wall track can be mounted on a wall, and the interface bracket can be coupled to the wall track.

(30) In some example configurations, the mount assembly 120A can include a display interface 118

and a support tray (e.g., the support tray **122A** of FIGS. 1-2). The display interface **118** and the support tray **122A** can be rotatably coupled to the arm assembly **110A** about a vertical axis **125**. In some example configurations, a rotation of the support tray **122A** can force a rotation of the display interface **118**, whereas the display interface **118** can be capable of rotating independent of the support tray **122A**. The display interface **118** can further include a support column **124** and a tilt assembly **126** coupled to the support column **124**.

(31) The tilt assembly **126** can include a display interface bracket. The display interface bracket can be adapted to be coupled to an electronic display **105**. The display interface **118** can be configured to hold an electronic display **105** above the support tray **122A**, and the display interface **118** can be configured to change an orientation of the electronic display **105** relative to the support tray **122A** (e.g., change a distance or angle of the electronic display **105** relative to the support tray **122A**).

(32) In some example configurations, a worksurface **121** can be coupled to the support tray **122A**. The user of the workstation **100** can place personal equipment (e.g., papers, a notebook, a calculator, one or more pens, or the like) on the worksurface **121**. In some example configurations, a keyboard can be placed over the worksurface **121**.

(33) In other example configurations, a slide out tray **128** (e.g., a keyboard tray) can be coupled to the support tray **122A** as illustrated in FIG. 2. The slide out tray **128** can be configured to hold an input device (e.g., a keyboard **106**) which can be placed over the slide out tray **128**. The slide out tray **128** can translate between a stowed orientation of where the keyboard **106** can be stowed under the worksurface **121** and an extended orientation where the slide out tray **128** can be pulled out (e.g., to locate it in front of the worksurface **121**) to make the keyboard **106** accessible as illustrated in FIG. 2. In some example configurations, the slide out tray **128** can be used as a drawer to store personal items.

(34) In some example configurations, a storage compartment **123** can be coupled to the support tray **122A** as illustrated in FIG. 1. The storage compartment **123** can be a separate component, and it can be secured to an underside of the support tray **122A** using one or more mechanical fasteners (screws, rivets, clamps, notches, latches, or the like). In other example configurations, the storage compartment **123** can be formed as part of the support tray **122A**. The storage compartment **123** can be used to store one or more electronic components including, but not limited to, a computer (e.g., a fat, thin, or zero client), a charging adaptor, one or more cables, or the like.

(35) FIG. 3 is a side view of a workstation **100** according to an example configuration of the current disclosure. The workstation **100** can include an arm assembly **110B**. The arm assembly **110B** can include a single arm (e.g., a height adjustable arm **113**) where the height adjustable arm **113** can be similar to the second arm **116** of FIG. 1. The arm assembly **110B** can have a first end **142** and a second end **144**. The arm assembly **110B** can be rotatably coupled to a structure interface **112** on the first end **142**, and a mount assembly **120B** can be rotatably coupled to the second end **144** of the arm assembly **110B** about a vertical axis **125**. The mount assembly **120B** can further include a support tray **122B** and a display interface **118**. In some example configurations, a rotation of the support tray **122A** can force a rotation of the display interface **118**, whereas the display interface **118** can be capable of rotating independent of the support tray **122A**.

(36) The display interface **118** can be configured to hold an electronic display **105** over the support tray **122B**. An input device (e.g., a keyboard **106**) can be located over the support tray **122B**. The arm assembly **110B** can be used to change an orientation (e.g., vertical orientation, horizontal orientation, or both) of the electronic display **105** and the keyboard **106** relative to the structure interface **112**.

(37) FIGS. 4A-4B are exploded views of a platform **210** and partial views of an arm assembly **110** according to some example configurations of the current disclosure. The platform **210** can be coupled to an end of an arm assembly **110** (e.g., the second end **144** of the arm assembly **110B** of FIG. 3). The platform **210** can be built in different configurations. In some example configurations, the platform **210A** can have a worksurface **121** coupled to the support tray **122A** as illustrated in

FIG. 4A. A keyboard tray (e.g., the slide-out tray **128** of FIG. 2) can be coupled to the support tray **122A** below the worksurface **121**. In other example configurations, the platform **210B** can be built without a worksurface as illustrated in FIG. 4B. A keyboard tray can be built as an integral part of the support tray **122B**. In either one of these example configurations, an interface assembly **320** can be coupled to the rear end of the platform **210** proximate to its center. The interface assembly **320** can be configured to quickly couple the platform **210** to the arm assembly **110**.

(38) FIG. 5 is an isometric view of the rear end of a platform **210** (e.g., the platform **210A** of FIG. 4A). The interface assembly **320** can include an interface bracket **340** and a carrier block **350**. The interface bracket **340** can be coupled to a first end **351** (shown in FIG. 8) of the carrier block **350**, and a platform tilt hinge **325** can be located on a second end **352** (shown in FIG. 8) of the carrier block **350**. The platform **210** can be rotatably coupled to the carrier block **350** at the platform tilt hinge **325**. The interface bracket **340** can be adapted to releasably couple the carrier block **350** to the arm assembly **110**.

(39) FIG. 6 is a side perspective view of a stowed orientation of the platform **210** (e.g., the platform **210A** of FIG. 4A) according to an example configuration of the current disclosure. The platform tilt hinge **325** can define a horizontal rotation axis **356** (shown in FIG. 8). The platform **210** can be rotated around the horizontal rotation axis **356** in a clockwise direction towards the electronic display **105** to place the platform in a stowed orientation (as illustrated in FIG. 6) when the workstation is not in use. In the stowed orientation, the footprint of the workstation **100** can be reduced to save office space. The platform **210** can be rotated around the horizontal rotation axis **356** in a counterclockwise direction away from the electronic display **105** to place the platform **210** in a use orientation when the workstation **100** is intended to be used by the user of the workstation.

(40) FIG. 7 is a rear isometric view of a platform **210** (e.g., the platform **210A** of FIG. 4A). The platform **210** can have a front end **211** and a rear end **212** opposite the front end **211**. In some example configurations, a recess **215** can be formed on the rear end **212** proximate to its center (e.g., the recess **215** can be formed on the support tray **122A**). The interface assembly **320** can be at least partially located inside the recess **215**. One or more apertures **217** can be formed on the recess. The platform **210** can be coupled to the interface assembly **320** through one or more apertures **217**.

(41) FIGS. 8-9 are isometric views showing interface assemblies (e.g., the interface assembly **320** of FIG. 5) in a folded orientation and in an extended orientation, respectively, according to example configurations of the current disclosure. The interface assembly **320** can include a carrier block **350**, an interface bracket **340**, and a hinge bracket **360**. In some example configurations, a slider **370** can be coupled to an upper surface **349** of the carrier block **350**.

(42) The hinge bracket **360** can have a hinge portion **361** and a flat portion **362**. The hinge portion **361** can be rounded. The flat portion **362** can extend from the hinge portion **361** in a transverse direction. One or more threaded holes **364** can be formed on the flat portion **362**. The one or more threaded holes **364** can overlap with the one or more apertures **217** when the platform **210** is coupled to the interface assembly **320**. One or more mechanical fasteners (e.g., screws, rivets, or the like) can be inserted through the one or more apertures **217** located on the support tray **122** and can engage with the one or more threaded holes **364** located on the hinge bracket **360** to securely couple the support tray **122** to the hinge bracket **360**.

(43) The hinge bracket **360** can be rotatably coupled to the carrier block **350** at the platform tilt hinge **325**. The hinge bracket **360** can be adapted to rotate between a folded orientation and an extended orientation. In the folded orientation, the hinge bracket **360** can be folded against the carrier block **350** as illustrated in FIG. 8. In the folded orientation, the hinge bracket **360** can be placed in a downward orientation (e.g., substantially vertical) and it can rest against the second end **352** of the carrier block **350**. The hinge bracket **360** can rotate (e.g., rotate 90 degrees, or the like) from the folded orientation to the extended orientation in clockwise direction around the horizontal rotation axis **356**. In the extended orientation, the hinge bracket **360** can be placed in a flat

orientation (e.g., substantially horizontal) as illustrated in FIG. 9. Since the support tray 122 is coupled to the hinge bracket 360, the support tray 122, and thus, the platform 210 can rotate with the hinge bracket 360 from the use orientation to the stowed orientation as the hinge bracket 360 rotates from the folded orientation to the extended orientation. The folded orientation of the hinge bracket 360 can correspond to the use orientation of the platform 210, and the extended orientation of the hinge bracket 360 can correspond to the stowed orientation of the platform 210.

(44) FIG. 10 is an isometric view of the carrier block 350 of FIG. 8. The carrier block 350 can have a first end 351 and a second end 352. A rear wall 353 can be built into the carrier block 350 proximate the first end 351. The carrier block 350 can further include a first side wall 354 and a second side wall 355. The first side wall 354 and the second side wall 355 can extend from the rear wall 353 in transverse direction from the first end 351 to the second end 352 of the carrier block 350. An upper surface 349 can be coupled to the first side wall 354, the second side wall 355 and the rear wall 353 of the carrier block 350 proximate an upper end 348 of the carrier block 350. The second end 352 and a lower end 347 opposite the upper end 348 of the carrier block 350 can be open between the first side wall 354 and the second side wall 355.

(45) In an example configuration, a first spacer 357 and a second spacer 358 can be coupled to the rear wall 353 as illustrated in FIG. 10. The first spacer 357 and the second spacer 358 can extend from the rear wall 353 in transverse direction towards the second end 352. The carrier block 350 can further include a first aperture 371 and a second aperture 372. The first aperture 371 and the second aperture 372 can be elongated holes formed on the upper surface 349.

(46) In an example configuration, a first through hole 344 and a second through hole 345 can be formed on the first side wall 354 and the second side wall 355, respectively. The first through hole 344 and the second through hole 345 can be located proximate to the upper end 348 and proximate to the second end 352. The first through hole 344 and the second through hole 345 can be coaxial. An axis passing through the centers of the first through hole 344 and the second through hole 345 can form a first rotation axis 3561 as illustrated in FIG. 10.

(47) FIG. 11 is an isometric view of the hinge bracket 360 of FIG. 8. The hinge bracket 360 can have a hinge portion 361 and a flat portion 362. A first side wall 365 and a second side wall 366 can be formed on the hinge bracket 360. The first side wall 365 and a second side wall 366 can extend from the flat portion 362 in transverse direction. The first side wall 365 and a second side wall 366 can be proximate to the hinge portion 361. A first through hole 367 and a second through hole 368 can be formed on the first side wall 365 and the second side wall 366, respectively. The first through hole 367 and the second through hole 345 can be coaxial. An axis passing through the centers of the first through hole 344 and the second through hole 345 can form a second rotation axis 3562 for the hinge bracket 360 as illustrated in FIG. 11.

(48) In an example configuration, a lock bracket 380 can be coupled to one of the first side wall 365 or the second side wall 366 of the hinge bracket 360 (e.g., coupled to the second side wall 366 as illustrated in FIG. 11). In other example configurations, the lock bracket 380 can be formed as an integral part of the hinge bracket 360. The lock bracket 380 can include a ramp 382 and a recess 384.

(49) FIG. 12 is an isometric view of a spring assembly 390 according to an example configuration of the current disclosure. The spring assembly 390 can include one or more torsion springs (e.g., a first torsion spring 392 and a second torsion spring 395). The first torsion spring 392 and the second torsion spring 395 can be made of a known spring wire. The first torsion spring 392 and the second torsion spring 395 can be concentric.

(50) In some example configurations, the spring wire can have a round cross-section or a square cross-section. One or more coils of the spring wire can be wound in circular orientation to form a body of the first torsion spring 392 as illustrated in FIG. 12. The first torsion spring 392 can have a first leg 393 and a second leg 394. The first leg 393 and the second leg 394 of the first torsion spring can extend away from the body of the first torsion spring 392. Similarly, one or more coils

of the spring wire can be wound in circular orientation to form a body of the second torsion spring **395**. The second torsion spring **395** can have a first leg **396** and a second leg **397**. The first leg **396** and the second leg **397** of the second torsion spring **395** can extend away from the body of the second torsion spring **395**. In some example configurations, the first leg **393** of the first torsion spring **392** and the first leg **396** of the second torsion spring **395** can be parallel to each other, and the second leg **394** of the first torsion spring **392** and the second leg **397** of the second torsion spring **395** can be parallel to each other.

(51) The spring assembly **390** can further include a mandrel **398**. The mandrel **398** can have a circular cross-section, and the mandrel **398** can have a through hole **399** proximate to its center. The mandrel **398** can be at least partially located inside the first torsion spring **392** and the second torsion spring **395**. In some example configurations, a mechanical fastener **391** (e.g., a screw, a rivet, or the like) can be inserted through the through hole **399** located at the center of the mandrel **398** to form a third rotation axis **3563** for the spring assembly **390** as illustrated in FIG. 12.

(52) FIG. 13 is an isometric view of the interface assembly **320** according to an example configuration of the current disclosure. In the interface assembly **320**, the hinge bracket **360** can be located between the first side wall **354** and second side wall **355** of the carrier block **350**, and the spring assembly **390** can be located between the first side wall **365** and the second side wall **366** of the hinge bracket **360**. The first rotation axis **3561**, the second rotation axis **3562**, and the third rotation axis **3563** can coincide with each other to form the horizontal rotation axis **356** (shown in FIG. 8). In an example configuration, the first leg **393** of the first torsion spring **392** can rest against the first spacer **357**, and the second leg **394** of the first torsion spring **392** can rest against the flat portion **362** of the hinge bracket **360**, similarly, the first leg **396** of the second torsion spring **395** can rest against the second spacer **358** and the second leg **397** of the second torsion spring **395** can rest against the flat portion **362** of the hinge bracket **360**. The second leg **394** of the first torsion spring **392** and the second leg **397** of the second torsion spring **395** can apply a force on to the flat portion **362** to bias the hinge bracket **360** towards the extended orientation (shown on FIG. 9). Since the platform **210** is coupled to the hinge bracket **360**, the first torsion spring **392** and the second torsion spring **395** can bias the platform **210** to rotate in a clockwise direction towards the stowed orientation shown in FIG. 6.

(53) FIG. 14 is an isometric view of an interface assembly **420** according to another example configuration of the current disclosure. The interface assembly **420** can include a carrier block **350** and a hinge bracket **460**. The hinge bracket **460** can be rotatably coupled to the carrier block **350** around a horizontal rotation axis **356** as discussed earlier in association with the FIG. 13.

(54) The hinge bracket **460** can include a flat portion **462**. In some example configurations, an adjustment assembly **400** can be coupled to the hinge bracket **460**. The adjustment assembly **400** can include a holding bracket **401** and an adjustment bracket **402**. The holding bracket **401** can be formed in a U-shaped configuration having a first side **403**, a second side **404** and a base **405**. The holding bracket **401** can be coupled to the flat portion **462** of the hinge bracket **460**. The base **405** of the holding bracket **401** can be parallel to the flat portion **462** of the hinge bracket **460**.

(55) The adjustment bracket **402** can be located between the first side **403** and the second side **404** of the holding bracket **401**. The adjustment bracket **402** can translate relative to the holding bracket **401** in a path perpendicular to the base **405**. An adjustment screw **406** can be coupled to the base **405** of the holding bracket **401** and coupled to the flat portion **462** of the hinge bracket **460**. The adjustment screw can be perpendicular to the base **405**. The adjustment screw **406** can be rotatably coupled to the holding bracket **401**. The adjustment screw **406** can have a screw head **407**. The screw head **407** can be located proximate to the base **405**. A nut **408** can be fixedly attached to the adjustment screw **406** proximate to the hinge bracket **460** as illustrated in FIG. 14. The screw head **407** and the nut **408** can prevent the adjustment screw **406** from translating perpendicular to the base **405**.

(56) A threaded hole can be formed on the adjustment bracket **402**. The threaded hole can be

concentric with the adjustment screw **406**. The adjustment screw **406** can be at least partially located inside the threaded hole located on the adjustment bracket **402**, and the adjustment screw **406** can be threadingly engaged with the adjustment bracket **402**. The adjustment bracket **402** can be adapted to translate along the longitudinal direction of the adjustment screw **406** when the adjustment screw **406** is rotated.

(57) In an example configuration, the second leg **394** of the first torsion spring **392** and the second leg **397** of the second torsion spring **395** can rest against the adjustment bracket **402**, as illustrated in FIG. **14**. Translating the adjustment bracket **402** by turning the adjustment screw **406** can change an angle between the first leg and the second leg of the first torsion spring **392** and the second torsion spring **395** (e.g., change an angle between the first leg **393** and the second leg **394** of the first torsion spring **392**). The angle between the first leg and the second leg of the torsion springs can be related to the spring tension.

(58) Manipulating the adjustment assembly **400** (e.g., turning the adjustment screw **406**), a user of the workstation **100** can selectively adjust a tension of the first torsion spring **392** and the second torsion spring **395** (e.g., turning the adjustment screw **406** in clockwise direction can increase the tension and turning the adjustment screw **406** in counterclockwise direction can decrease the tension). Increased tension of the torsion spring can apply an increased torque to the hinge bracket **360** in clockwise direction to counter a weight of the platform **210** (e.g., increase the tension of the torsion springs to counter heavier platform weights or decrease the tension of the torsion springs to counter lighter platform weights).

(59) FIGS. **15-16** are cross-sectional views of the interface assembly **320** of FIG. **9**. Front section of the interface assembly **320** is removed to make the internal components visible. The interface assembly **320** can include a carrier block **350** and a hinge bracket **360** as illustrated in FIG. **9**. The carrier block can have a rear wall **353**, an upper surface **349**, a first side wall **354**, and a second side wall **355** as illustrated in FIG. **10**. A first aperture **371** and a second aperture **372** can be formed on the upper surface **349**. The first aperture **371** and the second aperture **372** can be elongated in a first direction **410**. The first direction **410** can be perpendicular to the second side wall **355** and points away from the second side wall **355** as illustrated in FIG. **15**.

(60) In some example configurations, the interface assembly can include a lock assembly **375**. The lock assembly **375** can further include a slider **370**. The slider **370** can be located proximate to the upper surface **349**. A mechanical fastener **411** (e.g., a screw, a pin, or the like) can be inserted through the first aperture **371** and coupled to the slider **370**. A lock pin **412** can be inserted through the second aperture **372** and coupled to the slider **370**. The slider **370** can translate relative to the carrier block **350** in parallel to the first direction **410**. The mechanical fastener **411** and the lock pin **412** inserted through the first aperture **371** and the second aperture **372**, respectively, can guide the slider **370** through its translation relative to the carrier block **350**.

(61) In some example configurations, the lock pin **412** can be formed in an L-shape as illustrated in FIG. **15**. The lock pin **412** can have a first end **413** and a second end **414**. The first end **413** of the lock pin **412** can be coupled to the slider **370** and the second end **414** of the lock pin **412** can be elongated parallel to the first direction **410** (e.g., elongated perpendicular to the second side wall **355**). The second end **414** of the lock pin **412** can be formed in a profile (e.g., rounded, tapered, or the like).

(62) The interface assembly **320** can further include an energy storage member **416** (e.g., a tension spring, torsion spring, compression spring, or the like). The energy storage member **416** can be coupled between the lock pin **412** and the carrier block **350**. The energy storage member **416** can bias the slider **370** in a second direction opposite the first direction **410**.

(63) In the interface assembly **320**, the second side wall **366** of the hinge bracket **360** can be adjacent to the second side wall **355** of the carrier block **350**. A lock bracket **380** having a ramp **382** and a recess **384** can be coupled to the second side wall **366** of the hinge bracket **360** as illustrated in FIG. **11**. The ramp **382** and the recess **384** can translate in an arcuate path in front of the **414** as

the hinge bracket **360** rotates around the horizontal rotation axis **356** relative to the carrier block **350**.

(64) The hinge bracket **360**, and thus, the lock bracket **380** can be rotatably coupled with the carrier block **350**. As the hinge bracket **360** rotates from the extended orientation (shown in FIG. **9**) to the folded orientation (shown in FIG. **8**), the ramp **382** can contact the second end **414** of the lock pin **412** at a first angle of rotation (e.g., 80 degrees of rotation from the extended orientation, or the like). As the lock bracket **380** continues rotating together with the hinge bracket **360**, the ramp **382** can push the lock pin **412** and the slider **370** in the first direction **410** by stretching the energy storage member **416**. When the hinge bracket **360** can be placed in the folded orientation at a second angle of rotation (e.g., 90 degrees of rotation from the extended orientation, or the like), the recess **384** can coincide with the second end **414** of the lock pin **412**. In the folded orientation, the energy storage member **416** can pull the lock pin **412** and the slider **370** in the second direction opposite the first direction **410** such that the second end **414** of the lock pin **412** can be inserted into the recess **384** to lock the hinge bracket **360** in the folded orientation as illustrated in FIG. **16**.

(65) In some example configurations, the slider **370** can have a tab **374**. A user of the workstation **100** can interact with the tab **374** to push the slider **370** and the lock pin **412** in the first direction **410** to unlock the hinge bracket **360** (e.g., the second end **414** can move away from the recess **384** to disengage from the lock bracket **380** as illustrated in FIG. **15**). Once the lock bracket **380** is unlocked, the user of the workstation **100** can rotate the hinge bracket **360** from folded orientation to the extended orientation to put the platform **210** into a stowed orientation as illustrated in FIG. **6**. The one or more energy storage members (e.g., the first torsion spring **392** and the second torsion spring **395**, or the like) can provide lift assist to reduce the amount of force that can be exerted by the user to rotate the platform **210** from extended orientation to the stowed orientation.

(66) In some example configurations, the one or more energy storage members (e.g., the first torsion spring **392** and the second torsion spring **395**) can have enough tension to counter the weight of the platform **210** and automatically rotate the platform **210** at least part of the way from the extended orientation towards the stowed orientation when the lock pin **412** is disengaged from the lock bracket **380**. The one or more torsion springs can apply a force on to the hinge bracket **360** to rotate it from folded orientation towards the extended orientation. When the user of the workstation **100** unlocks the hinge bracket **360** (e.g., pushes the slider **370** in the first direction **410** to disengage the lock pin **412** from the lock bracket **380** by removing the second end **414** of the lock pin **412** from the recess **384**), the force applied to the hinge bracket **360** by the one or more torsion springs can rotate the hinge bracket **360** automatically to a third angle of rotation (e.g., 85 degrees of rotation from the extended orientation, or the like). This initial automatic rotation can indicate to the user that the hinge bracket **360** is unlocked and the platform **210** is ready to be rotated to the stowed orientation.

(67) FIG. **17** is a side view of the second arm **116** of FIG. **1**. The second arm **116** can include a first block **450**, a second block **451**, an upper link arm **455** and a lower link arm **456**. The upper link arm **455** can be rotatably coupled to the first block **450** on one end and rotatably coupled to the second block **451** on the other end. The lower link arm **456** can be rotatably coupled to the first block **450** on one end and rotatably coupled to the second block **451** on the other end. The upper link arm **455** and the lower link arm **456** can be parallel to each other to form a parallel linkage mechanism. The upper link arm **455** and the lower link arm **456** can rotate relative to the first block **450** to adjust a vertical distance between the first block **450** and the second block **451**. The upper link arm **455** and the lower link arm **456** can stay parallel during the adjustment to keep the second block **451** leveled at any position.

(68) In some example configurations, the first block **450** can be used to mount the second arm **116** to a structure (e.g., mount to the first arm **114**, mount to the structure interface **112**, or the like). One or more components (e.g., the platform **210**, the display interface **118**, or the like) can be coupled to the second block **451**. The second arm **116** can be configured to translate the one or

more components relative to the first block **450** when the upper link arm **455** and lower link arm **456** rotate relative to the **450**.

(69) FIG. **18** is a cross-sectional view of the second arm of FIG. **17**. In some example configurations, the second arm **116** can include an arm counterbalance mechanism **465** to provide lift assist during the height adjustment. The arm counterbalance mechanism **465** can provide a lift force to counter at least a portion of the combined weight (e.g., combined weight of the platform **210**, electronic display **105**, or the like) attached to the second arm **116**.

(70) The arm counterbalance mechanism **465** can include an energy storage member **466** (e.g., a compression spring, or the like). The energy storage member **466** can be coupled to the lower link arm **456**. The energy storage member **466** can have a first end and a second end. A cam follower **467** can be coupled to the first end of the energy storage member **466**, and a tension adjustment assembly **470** can be coupled to the second end of the energy storage member **466**. The cam follower **467** can be in contact with a cam **453** formed on the first block **450** throughout the rotation of the lower link arm **456** relative to the first block **450**.

(71) FIG. **19** is an enlarged cross-sectional view of the tension adjustment assembly **470** of FIG. **18**. The tension adjustment assembly **470** can be coupled to the lower link arm **456**. The tension adjustment assembly **470** can include a first bevel gear **471** and a second bevel gear **472**. The first bevel gear **471** can rotate around a first axis **473**, and the second bevel gear can rotate around a second axis **474**. The second axis **474** can be perpendicular to the first axis **473**. The second bevel gear **472** can engage (e.g., interlock, join, contact, or the like) with the first bevel gear **471** such that a rotation of the first bevel gear **471** can cause a rotation of the second bevel gear **472**.

(72) The tension adjustment assembly **470** can also include an adjustment screw **475** and an adjustment bracket **476**. The adjustment screw **475** can be concentric with the second bevel gear **472**. The adjustment screw **475** can be coupled to the second bevel gear **472** such that the adjustment screw **475** can be adapted to rotate together with the second bevel gear **472** in unison around the second axis **474**.

(73) The adjustment bracket **476** can have a threaded hole **4761** proximate to its center. The adjustment screw **475** can be at least partially located inside the threaded hole. The adjustment screw **475** can be threadingly engaged with the adjustment bracket **476**. The adjustment bracket **476** can be configured to translate along the axial direction of the adjustment screw **475** (e.g., along the second axis **474**) as the adjustment screw **475** is rotated. The second end of the energy storage member **466** can be coupled to (e.g., presses against, or the like) the adjustment bracket **476** as illustrated in FIG. **19**. The adjustment bracket **476** can be configured to change a tension (or compression) of the energy storage member **466** as the adjustment bracket translates along the second axis **474**.

(74) An elevation of the second block **451** relative to the first block **450** can be changed between a raised position and a lowered position by rotating the lower link arm **456** relative to the first block **450**. The energy storage member **466** can bias the second block **451** towards the raised position. During the rotation of the second arm **116**, the cam follower **467** can follow a cam profile of the cam **453** to further increase the tension (or compression) of the energy storage member **466**. The cam profile of the cam **453** can be configured such that the lift force provided by the arm counterbalance mechanism **465** can be substantially constant throughout the rotation of the lower link arm **456** despite the increased tension (or compression) of the energy storage member **466**.

(75) FIG. **20** is an exploded view of the tension adjustment assembly **470** according to an example configuration of the current disclosure. The first bevel gear **471** can be formed on a first bevel gear shaft **480** and the second bevel gear **472** can be formed on a second bevel gear shaft **490**. The first bevel gear shaft **480** can cooperate with the second bevel gear shaft **490** to drive the adjustment screw **475** which then translates the adjustment bracket **476** along the second axis **474** to increase the tension of the energy storage member **466**.

(76) The first bevel gear shaft **480** can have a first end **481** and a second end **482** opposite the first

end **481**. The first bevel gear **471** can be formed on the first bevel gear shaft **480** proximate to the first end **481**. In some configurations, the first bevel gear **471** can be formed as an integral part of the first bevel gear shaft **480**. In other configurations, the first bevel gear **471** can be formed separately and fixedly attached to the first bevel gear shaft **480** during assembly.

(77) The first bevel gear shaft **480** can be formed in four sections including a first section **483**, a second section **484**, a third section **485**, and a fourth section **486**. The first section **483** can extend from the first bevel gear **471** in one direction towards the first end **481**. The first section **483** can be formed in a circular cross-section, and it can have a first diameter. The second section **484** can extend from the first bevel gear **471** in the other direction towards the second end **482**. The third section **485** can extend from the second section **484** towards the second end **482**, and the fourth section **486** can extend from the third section **485** towards the second end **482**. The second section **484**, the third section **485**, and the fourth section **486** can be formed in circular cross-sections, and they can have a second diameter, a third diameter, and a fourth diameter, respectively. The third diameter of the third section **485** can be smaller than the second diameter of the second section **484** and the fourth diameter of the fourth section **486**. A first face **487** can be formed at the intersection between the second section **484** and the third section **485**, and a second face **488** can be formed at the intersection between the third section **485** and the fourth section **486**. The first face **487** and the second face **488** can extend from the third section in transverse direction. The third section **485** can have a length **4851** between the first face **487** and the second face **488**. A hexagonal shaped recess **489** (shown in FIG. 19) can be formed at the second end **482** of the first bevel gear shaft **480**.

(78) The second bevel gear shaft **490** can have a first end **491** and a second end **492**. The second bevel gear **472** can be formed on the second bevel gear shaft **490** proximate to the first end **491**. In some configurations, the second bevel gear **472** can be formed as an integral part of the second bevel gear shaft **490**. In other configurations, the second bevel gear **472** can be formed separately and fixedly attached to the second bevel gear shaft **490** during assembly.

(79) The second bevel gear shaft **490** can be formed in three sections including a first section **493**, a second section **494**, and a third section **495**. The first section **493** can extend from the second bevel gear **472** in one direction towards the first end **491**. The first section **493** can be formed in a circular cross-section, and it can have a first diameter **4931**. The second section **494** can extend from the second bevel gear **472** in the other direction towards the second end **492**. The third section **495** can extend from the second section **494** towards the second end **492**. The second section **494**, and the third section **495** can be formed in circular cross-sections, and they can have a second diameter, and a third diameter, respectively. The third diameter of the third section **495** can be larger than the second diameter of the second section **494**. A recess **496** (shown in FIG. 19) can be formed at the second end **492** of the second bevel gear shaft **490**. The recess **496** can have any cross-section including, but not limited to, hexagonal, square, or the like.

(80) In some example configurations, the length **4851** of the third section **485** of the first bevel gear shaft **480** can be configured to receive the first diameter **4931** of the first section **493** of the second bevel gear shaft **490**.

(81) The adjustment screw **475** can have a first end **477** and a second end **478** opposite the first end **477**. A boss **479** can be formed at the first end **477**. A cross-section of the boss **479** can be formed in any shape including, but not limited to, hexagonal, square, or the like. The cross-section of the boss **479** can match the cross-section of the recess **496** located on the second bevel gear shaft **490**. In some configurations, the boss **479** can be inserted into the hexagonal shaped recess **496** to couple the adjustment screw **475** with the second bevel gear shaft **490** such that the adjustment screw **475** can be configured to rotate together with the second bevel gear shaft **490**.

(82) The adjustment screw **475** can extend from the boss **479** towards the second end to form a threaded shaft. An adjustment bracket **476** can threadingly engage with the adjustment screw **475** over the threaded shaft. The adjustment screw **475** can be flattened proximate to the second end **478** to help limit the adjustment bracket **476** from accidentally disengaging from the adjustment

screw **475**.

(83) The first section **493** of the second bevel gear shaft **490** can be located proximate the third section **485** of the first bevel gear shaft **480**. The length **4851** of the third section **485** can be configured (e.g., the length **4851** can be larger than the first diameter **4931** of the first section **493**) such that the first section **493** can be received between the first face **487** and the second face **488**. The first face **487** and the second face **488** can prevent the second bevel gear shaft **490** from translating in a direction parallel to the first axis **473**.

(84) Going back to FIG. **20**, a hub **497** can be coupled to the adjustment screw **475** proximate to the first end **477**. The hub **497** can house one or more thrust washers **498** shown in FIG. **19**. In some configurations, the hub **497** can be formed as an integral part of the adjustment screw **475**, or in other configurations, the hub **497** can be formed separately and coupled to the adjustment screw **475** during assembly.

(85) FIG. **21** is a bottom isometric view of the lower link arm **456** according to an example configuration of the current disclosure. The lower link arm **456** can be formed in a U-shaped cross-section having a first side wall **501**, a second side wall **502** opposite the first side wall **501** and a bottom wall **503**. The bottom wall **503** can be coupled to the first side wall **501** and the second side wall **502**. In some configurations, an internal wall **504** can be formed proximate to an end of the lower link arm **456**. The internal wall **504** can extend from the bottom wall **503** and it can be coupled to the first side wall **501** and the second side wall **502**. The internal wall **504** can extend in transverse direction from the bottom wall **503** through at least a portion of the height of the first side wall **501** and the second side wall **502**. A circular opening **505** can be formed on the internal wall **504** proximate to its center. A radius of the circular opening can match the radius of the second section **494** of the second bevel gear shaft **490**.

(86) The lower link arm **456** can further include one or more bosses **506** and an aperture **507**. The aperture **507** can be formed on the bottom wall **503** proximate to the internal wall **504**. The one or more bosses **506** and the aperture **507** can be used to couple the tension adjustment assembly **470** to the lower link arm **456**.

(87) FIG. **22** is a housing **510** according to an example configuration of the current disclosure. The housing **510** can be coupled to the lower link arm **456** and it can be configured to hold the first bevel gear shaft **480** and the second bevel gear shaft **490**. The housing **510** can have a center opening **511** and one or more through holes **512** located on the side of the center opening **511**. The diameter of the center opening **511** can be configured to receive at least the fourth section **486** of the first bevel gear shaft **480**.

(88) The housing **510** can further include a first flange **513**, a second flange **514**, and a third flange **515** as illustrated in FIG. **22**. The first flange **513** can be adjacent to the center opening **511**. A portion of the wall of the center opening **511** can be removed above the first flange **513**. The second flange **514** can be located away from the first flange **513**. The distance between the first flange **513** and the second flange **514** can be configured to receive the second bevel gear **472**. The third flange **515** can be adjacent to the second flange **514**.

(89) In some example configurations, the first flange **513**, the second flange **514**, and the third flange **515** can have concave round profiles at their respective upper ends. A radius of the concave profile on the first flange **513** can match the radius of the first section **493** of the second bevel gear shaft **490**. A radius of the concave profile on the second flange **514** can match the radius of the second section **494** of the second bevel gear shaft **490**. A radius of the concave profile on the third flange **515** can match the radius of the third section **495** of the second bevel gear shaft **490**.

(90) FIG. **23** is an isometric view of a gear assembly **520** according to an example configuration of the current disclosure. The gear assembly **520** can include the housing **510**, the first bevel gear shaft **480** having the first bevel gear **471** and the second bevel gear shaft **490** having the second bevel gear **472**. The third section **485** and the fourth section **486** of the first bevel gear shaft **480** can be inserted into the center opening **511** of the housing **510**. The second section **484** can extend

outside the housing **510**. In some configurations, the first face **487** can be seated over an upper end **516** of the housing **510**.

(91) The second bevel gear shaft **490** can be inserted into the housing **510** such that the second bevel gear **472** can be located between the first flange **513** and the second flange **514**. The first section **493** of the second bevel gear shaft **490** can be seated over the first flange **513**, the second section **494** of the second bevel gear shaft **490** can be seated over the second flange **514**, and the third section **495** of the second bevel gear shaft **490** can be seated over the third flange **515**.

(92) The gear assembly **520** of FIG. **23** can be coupled to the lower link arm **456**. The first section **483** of the first bevel gear shaft **480** can be inserted into the aperture **507**, and the one or more through holes **512** can overlap with the one or more bosses **506**. One or more mechanical fasteners (e.g., screws, pins, or the like) can be inserted through the one or more through holes **512** and threadingly engage with the one or more bosses **506** to secure the housing to the lower link arm **456**. During the attachment of the gear assembly **520** to the lower link arm **456**, the internal wall **504** can be inserted between the second bevel gear **472** and the third section **495** of the second bevel gear shaft **490**. The circular opening **505** can be seated over the second section **494** of the second bevel gear shaft **490** to trap it between the circular opening **505** and the second flange **514**.

(93) Going back to FIG. **19**, a tool (e.g., a hex wrench, or the like) can be inserted into the hexagonal shaped recess **489** to engage with the first bevel gear shaft **480**. The tool can be manipulated (e.g., rotated, or the like) to rotate the first bevel gear **471** and the second bevel gear **472**, and thus, to rotate the adjustment screw **475**. A rotation of the adjustment screw **475** can translate the adjustment bracket **476** to adjust the tension of the energy storage member **466** as discussed above.

(94) Each of these non-limiting examples can stand on its own or can be combined in any permutation or combination with any one or more of the other examples.

(95) The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the present subject matter can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described.

However, the present inventor also contemplates examples in which only those elements shown or described are provided. Moreover, the present inventor also contemplates examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

(96) In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls.

(97) In the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

(98) The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the

Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the present subject matter should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Additional Notes and Aspects

(99) Aspect 1 may include or use subject matter (such as an apparatus, a system, a device, a method, a means for performing acts, or a device readable medium including instructions that, when performed by the device, may cause the device to perform acts), such as may include or use a workstation support assembly comprising: an arm assembly having a first end rotatably coupled to a structure and a second end; and a mount assembly comprising: a keyboard support assembly adapted to support a keyboard; and a display support assembly adapted to support an electronic display over the keyboard support assembly, the display support assembly rotatably coupled with the arm assembly about a vertical axis proximate the second end; wherein the arm assembly is configured to translate the mount assembly relative to the structure.

(100) Aspect 2 may include or use, or may optionally be combined with the subject matter of Aspect 1, to optionally include or use, wherein the arm assembly comprises a first arm and a second arm, wherein the first arm is rotatably coupled to the structure on one end and rotatably coupled to the second arm on an opposite end of the first arm, wherein the second arm is rotatably coupled to the first arm on one end and rotatably coupled to the mount assembly on an opposite end of the second arm, wherein the first arm and the second arm are configured to cooperate to translate the mount assembly in a horizontal direction relative to the structure, and wherein the second arm is configured to translate the mount assembly in a vertical direction relative to the structure.

(101) Aspect 3 may include or use, or may optionally be combined with the subject matter of Aspect 2, to optionally include or use, wherein the second arm comprises: a first block coupled to the first arm; a second block coupled to the mount assembly; a first link arm rotatably coupled to the first block on one end and rotatably coupled to the second block on an opposite end; and a second link arm rotatably coupled to the first block on one end and rotatably coupled to the second block on an opposite end; wherein the first block, the second block, the first link arm, and the second link arm are configured to cooperate to translate the mount assembly in a vertical direction relative to the structure.

(102) Aspect 4 may include or use, or may optionally be combined with the subject matter of Aspect 3, to optionally include or use, wherein the second arm comprises a counterbalance mechanism, wherein the counterbalance mechanism is configured to generate a lift force to counter a weight of the mount assembly.

(103) Aspect 5 may include or use, or may optionally be combined with the subject matter of Aspect 4, to optionally include or use, wherein the counterbalance mechanism comprises: a cam having a cam profile; wherein the cam is coupled to the first block, an energy storage member having an energy storage member first end and an energy storage member second end; a cam follower coupled to the energy storage member first end; and an adjustment assembly coupled to the energy storage member second end; wherein the energy storage member, the cam follower, and the adjustment assembly are coupled to a lower link arm, wherein the cam follower is adapted to move over the cam profile as the second link arm rotates relative to the second block, and wherein the adjustment assembly is configured to change a tension of the energy storage member.

(104) Aspect 6 may include or use, or may optionally be combined with the subject matter of Aspect 5, to optionally include or use, wherein the adjustment assembly comprises: a first gear shaft having a first bevel gear; wherein the first gear shaft and the first bevel gear are configured to rotate about a first axis; a second gear shaft having a second bevel gear, wherein the second gear shaft and the second bevel gear are configured to rotate about a second axis, wherein the second

axis is perpendicular to the first axis, and an adjustment screw coupled to the second gear shaft, wherein the adjustment screw is coaxial with the second gear shaft at the second axis, and wherein the first bevel gear cooperates with the second bevel gear to rotate the adjustment screw about the second axis.

(105) Aspect 7 may include or use, or may optionally be combined with the subject matter of Aspect 6, to optionally include or use, wherein the first gear shaft is elongated between a first gear shaft first end and a first gear shaft second end, wherein the first bevel gear is coupled to the first gear shaft proximate the first gear shaft first end, the first gear shaft further comprises a recessed section between the first gear shaft first end and the first gear shaft second end, wherein the recessed section is formed in a round cross-section having a first length between a first face and a second face, wherein the first face and the second face extend from the recessed section in transverse direction.

(106) Aspect 8 may include or use, or may optionally be combined with the subject matter of Aspect 7, to optionally include or use, wherein the second gear shaft is elongated between a second gear shaft first end and a second gear shaft second end, wherein the second bevel gear is coupled to the second gear shaft proximate the second gear shaft first end, wherein the second gear shaft has a first diameter at the second gear shaft first end, wherein the first diameter is smaller than the first length, wherein the second gear shaft first end is configured to be inserted into the recessed section of the first gear shaft between the first face and the second face, wherein the first face and the second face are configured to prevent the second gear shaft to move in a direction parallel to the first axis, and wherein the adjustment screw is coupled to the second gear shaft.

(107) Aspect 9 may include or use, or may optionally be combined with the subject matter of Aspect 8, to optionally include or use, wherein the adjustment assembly comprises an adjustment bracket having a threaded hole at its center, wherein the adjustment bracket is threadingly engaged with the adjustment screw at the threaded hole, wherein the energy storage member second end is coupled to the adjustment bracket, and wherein the adjustment bracket is configured to translate along the adjustment screw as the adjustment screw is rotated to change the tension of the energy storage member.

(108) Aspect 10 may include or use, or may optionally be combined with the subject matter of Aspect 1, to optionally include or use, wherein the keyboard support assembly comprises: a support tray, and an interface assembly, wherein the interface assembly is coupled between the support tray and the arm assembly, and wherein the support tray is configured to rotate around a horizontal axis relative to the interface assembly between a use orientation where the support tray is substantially horizontal and a stowed orientation where the support tray is substantially vertical.

(109) Aspect 11 may include or use, or may optionally be combined with the subject matter of Aspect 10, to optionally include or use, wherein the interface assembly comprises: an interface bracket configured to be coupled to the arm assembly; a carrier block; and a hinge bracket configured to be coupled to the support tray; wherein the carrier block is coupled to the interface bracket on one end and coupled to the hinge bracket on an opposite end; wherein the hinge bracket is configured to rotate around the horizontal axis between a folded orientation where the hinge bracket is substantially vertical and an extended orientation where hinge bracket is substantially horizontal; and wherein the folded orientation of the hinge bracket corresponds to the use orientation of the support tray and the extended orientation of the hinge bracket corresponds to the stowed orientation of the support tray.

(110) Aspect 12 may include or use, or may optionally be combined with the subject matter of Aspect 11, to optionally include or use, wherein the interface assembly comprises a spring assembly having one or more torsion springs, the one or more torsion springs include a first leg and a second leg, wherein the first leg is coupled to the carrier block, and the second leg is coupled to the hinge bracket, and wherein the one or more torsion springs biases the hinge bracket towards the extended orientation.

(111) Aspect 13 may include or use, or may optionally be combined with the subject matter of Aspect 12, to optionally include or use, wherein the interface assembly comprises a torsion spring adjustment assembly, wherein the torsion spring adjustment assembly is coupled to the hinge bracket, the torsion spring adjustment assembly includes: a holding bracket coupled to the hinge bracket; a torsion spring adjustment bracket; and a torsion spring adjustment screw coupled between the holding bracket and the hinge bracket, wherein the torsion spring adjustment screw is threadingly engaged with the torsion spring adjustment bracket, wherein the second legs of the one or more torsion springs are coupled to the torsion spring adjustment bracket, and wherein a rotation of the torsion spring adjustment screw translates the torsion spring adjustment bracket to change a tension of the one or more torsion springs.

(112) Aspect 14 may include or use, or may optionally be combined with the subject matter of Aspect 12, to optionally include or use, wherein the interface assembly comprises a lock assembly including: a slider slidingly engaged with the carrier block; a lock pin coupled to the slider; and a spring coupled between the slider and the carrier block; wherein the slider translates between a locked configuration and an unlocked configuration relative to the carrier block, wherein the lock pin is configured to engage with the hinge bracket in the locked configuration when the support tray is in the use orientation, and disengage from the hinge bracket in the unlocked configuration allowing the support tray to be rotated to the stowed orientation, and wherein the spring is configured to bias the lock assembly towards the locked configuration.

(113) Aspect 15 may include or use, or may optionally be combined with the subject matter of Aspect 14, to optionally include or use, wherein the spring assembly is configured to automatically rotate the support tray a first angle from the use orientation towards the stowed orientation when the lock assembly is put in the unlocked configuration, and wherein the first angle is less than a full rotation angle between the use orientation and the stowed orientation.

(114) Aspect 16 may include or use, or may optionally be combined with the subject matter of Aspect 11, to optionally include or use, wherein the carrier block is rotatingly coupled to the interface bracket to level the support tray relative to the structure.

(115) Aspect 17 may include or use subject matter (such as an apparatus, a system, a device, a method, a means for performing acts, or a device readable medium including instructions that, when performed by the device, may cause the device to perform acts), such as may include or use a workstation support assembly comprising: a mount assembly comprising: a keyboard support adapted to support a keyboard; and a display support adapted to support an electronic display over the keyboard support an arm assembly configured to translate the mount assembly relative to a structure, the arm assembly including a counterbalance mechanism configured to generate a lift force to counter a weight of the mount assembly, the counterbalance mechanism comprising: an energy storage member; a cam defining a cam profile and a cam follower coupled to the energy storage member, the cam follower adapted to move over the cam profile as the second arm rotates; and an adjustment assembly coupled to the energy storage member, the adjustment assembly configured to adjust a tension of the energy storage member, the adjustment assembly comprising: an adjustment screw; a first gear shaft including a first bevel gear, the first gear shaft and the first bevel gear together configured to rotate about a first axis; and a second gear shaft including a second bevel gear, the second gear shaft and the second bevel gear configured to rotate about a second axis that is perpendicular to the first axis, the second gear shaft coupled to, and coaxial with, the adjustment screw, the first bevel gear configured to cooperate with the second bevel gear to rotate the adjustment screw about the second axis to adjust the tension of the energy storage member.

(116) Aspect 18 may include or use, or may optionally be combined with the subject matter of Aspect 17, to optionally include or use, wherein the first gear shaft is elongated between a first gear shaft first end and a first gear shaft second end, wherein the first bevel gear is coupled to the first gear shaft proximate the first gear shaft first end, the first gear shaft further comprises a recessed

section between the first gear shaft first end and the first gear shaft second end, wherein the recessed section is formed in a round cross-section having a first length between a first face and a second face, wherein the first face and the second face project from the recessed section in transverse direction.

(117) Aspect 19 may include or use, or may optionally be combined with the subject matter of Aspect 18, to optionally include or use, wherein the second gear shaft is elongated between a second gear shaft first end and a second gear shaft second end, wherein the second bevel gear is coupled to the second gear shaft proximate the second gear shaft first end, the second gear shaft has a first diameter at the second gear shaft first end, wherein the first diameter is smaller than the first length, wherein the second gear shaft first end is configured to be inserted into the recessed section of the first gear shaft between the first face and the second face, wherein the first face and the second face are configured to prevent the second gear shaft to move in a direction parallel to the first axis, and wherein the adjustment screw is coupled to the second gear shaft.

(118) Aspect 20 may include or use subject matter (such as an apparatus, a system, a device, a method, a means for performing acts, or a device readable medium including instructions that, when performed by the device, may cause the device to perform acts), such as may include or use a workstation support assembly comprising: an arm assembly having a first end rotatably coupled to a structure and a second end; and a mount assembly coupled to the second end, the mount assembly comprising: a display support adapted to support an electronic display; and a keyboard support adapted to support a keyboard, the keyboard support comprising: a support tray; and an interface assembly comprising: a carrier block configured to be coupled to the arm assembly; a hinge bracket coupled to the support tray and coupled to the carrier block, wherein the hinge bracket is configured to rotate about a horizontal axis relative to the carrier block; a spring assembly having one or more torsion springs, the one or more torsion springs include a first leg coupled to the carrier block, and a second leg coupled to the hinge bracket; and a lock assembly including: a lock pin; a slider coupled to the lock pin, wherein the slider translates between a locked configuration and an unlocked configuration relative to the carrier block; and a spring coupled between the slider and the carrier block, wherein the spring is configured to bias the slider towards the locked configuration; wherein the support tray is configured to rotate around the horizontal axis relative to the carrier block between a use orientation where the support tray is substantially horizontal and a stowed orientation where the support tray is substantially vertical; wherein the lock pin is configured to engage with the hinge bracket in the locked configuration when the support tray is in the use orientation, and disengage from the hinge bracket in the unlocked configuration allowing the support tray to be rotated to the stowed orientation; and wherein the spring assembly is configured to automatically rotate the support tray a first angle from the use orientation towards the stowed orientation in the unlocked configuration, wherein the first angle is less than a full rotation angle between the use orientation and the stowed orientation.

(119) Each of these non-limiting examples can stand on its own or can be combined in any permutation or combination with any one or more of the other examples.

(120) The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the present subject matter can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventor also contemplates examples in which only those elements shown or described are provided. Moreover, the present inventor also contemplates examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

(121) In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls.

(122) In the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

(123) The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the present subject matter should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A workstation support assembly comprising: a mount assembly comprising: a keyboard support adapted to support a keyboard; and a display support adapted to support an electronic display over the keyboard support; and an arm assembly configured to translate the mount assembly relative to a structure, the arm assembly including a counterbalance mechanism configured to generate a lift force to counter a weight of the mount assembly and the electronic display, the counterbalance mechanism comprising: an energy storage member; a cam defining a cam profile; a cam follower coupled to the energy storage member, the cam follower adapted to move over the cam profile as the arm assembly rotates; and an adjustment assembly coupled to the energy storage member, the adjustment assembly configured to adjust a tension of the energy storage member to adjust the lift force generated by the counterbalance mechanism, wherein the adjustment assembly comprises: an adjustment screw; a first gear assembly including: a first gear shaft; and a first bevel gear, the first gear shaft and the first bevel gear together configured to rotate about a first axis; and a second gear assembly including: a second gear shaft; and a second bevel gear, the second gear shaft and the second bevel gear configured to rotate about a second axis that is perpendicular to the first axis, the second gear shaft coupled to, and coaxial with, the adjustment screw, the first bevel gear configured to cooperate with the second bevel gear to rotate the adjustment screw about the second axis to adjust the tension of the energy storage member to adjust the lift force generated by the counterbalance mechanism.
2. The workstation support assembly of claim 1, wherein the first gear shaft is elongated between a first gear shaft first end and a first gear shaft second end, and wherein the first bevel gear is coupled to the first gear shaft proximate the first gear shaft first end.
3. The workstation support assembly of claim 2, wherein the first gear shaft further comprises: a recessed section between the first gear shaft first end and the first gear shaft second end, wherein: the recessed section is formed in a round cross-section; and each of a first face and a second face projects from the recessed section in transverse direction.
4. The workstation support assembly of claim 3, wherein the second gear shaft is elongated between a second gear shaft first end and a second gear shaft second end, and wherein the second

bevel gear is coupled to the second gear shaft proximate the second gear shaft first end.

5. The workstation support assembly of claim 4, wherein the second gear shaft has a first diameter at the second gear shaft first end, wherein the round cross-section has a first length between the first face and the second face.

6. The workstation support assembly of claim 5, wherein first diameter is smaller than the first length.

7. The workstation support assembly of claim 5, wherein the second gear shaft first end is insertable into the recessed section of the first gear shaft between the first face and the second face, wherein the first face and the second face are configured to limit movement of the second gear shaft in a direction parallel to the first axis, and wherein the adjustment screw is coupled to the second gear shaft.

8. The workstation support assembly of claim 1, wherein the cam profile is configured to maintain a substantially constant lift force throughout a range of motion of the arm assembly.

9. The workstation support assembly of claim 1, wherein the energy storage member comprises a compression spring, and wherein the adjustment assembly is operable to adjust a tension of the compression spring to vary the lift force provided by the counterbalance mechanism.

10. The workstation support assembly of claim 1, wherein the adjustment assembly is accessible via an external interface to allow for manual adjustment of the tension of the energy storage member without disassembling the arm assembly.

11. The workstation support assembly of claim 1, wherein the second gear shaft includes a recessed section configured to receive a tool for rotating the second gear shaft, thereby adjusting the tension of the energy storage member via the adjustment screw.

12. The workstation support assembly of claim 11, wherein the adjustment assembly further comprises: an adjustment bracket threadedly engaged with the adjustment screw, a position of the adjustment bracket along the adjustment screw determining the tension of the energy storage member.

13. The workstation support assembly of claim 12, wherein the adjustment bracket is coupled to a second end of the energy storage member, and movement of the adjustment bracket along the adjustment screw alters a force exerted by the energy storage member on the arm assembly.

14. The workstation support assembly of claim 13, wherein the arm assembly comprises: a lower link arm; and an upper link arm, the lower link arm and the upper link arm together form a parallel linkage mechanism.

15. The workstation support assembly of claim 14, wherein the counterbalance mechanism is operatively connected to the lower link arm to assist in vertical movement of the mount assembly.

16. The workstation support assembly of claim 15, wherein the lower link arm comprises a housing configured to contain the adjustment assembly, the housing attachable to the lower link arm to maintain alignment of the first bevel gear and the second bevel gear during operation.

17. The workstation support assembly of claim 16, wherein the housing includes a flange system configured to support the second gear shaft and to maintain the alignment of the second bevel gear with the first bevel gear.

18. The workstation support assembly of claim 17, wherein the flange system of the housing is configured to facilitate the arm assembly and maintenance of the adjustment assembly by providing access points for insertion and removal of the first gear shaft and the second gear shaft.

19. The workstation support assembly of claim 16, wherein adjustment screw extends through the housing to permit adjust the tension of the energy storage member to adjust the lift force generated by the counterbalance mechanism without disassembling the workstation support assembly.
